

Hospital Chatbot for Color Coded Clinical Pathways

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The Nuance of Human Conversation



"Human communication is inherently complex, unstructured, and contextual."

Natural Language Processing (NLP) & NLU

- **The Core Foundation:** NLP is the foundational language engine of the hospital chatbot.
- **Sub-branch:** We utilize Natural Language Understanding (NLU) combined with Text Classification.
- **Natural Language:** The messy, unstructured way humans communicate (e.g., subjective patient symptom descriptions).
- **Processing Phase:** Human text is translated into rigid mathematical structures (matrices and vectors). This allows a computer to read, process, and analyze the subjective input objectively.

Aligning NLP with Clinical Standards

- **Clinical Standard:** The South African Triage Scale (SATS) forms the backbone of our pathways.
- **Objective Validation:** The Triage Early Warning Score (TEWS) provides a deterministic, mathematical safety net based on vitals.
- **The Chatbot's Role:** By combining NLP with TEWS, the system effectively translates subjective, unstructured human complaints into objective mathematical scores that align directly with established SATS triage pathways.

Basic Machine Learning Concepts

Types of Learning

- **Supervised Learning:** Uses labeled data to teach the algorithm using known examples (e.g., historical triage records).
- **Unsupervised Learning:** Discovers hidden patterns independently in unlabeled data.

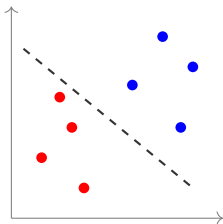
Examples: K-Nearest Neighbors, Support Vector Machines (SVM), Decision Trees, and **Neural Networks**.

Types of Models

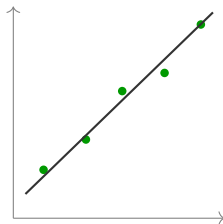
- **Classification Models:** Predict distinct categories (e.g., SATS color tiers).
- **Regression Models:** Predict continuous numerical values.

Visualizing Machine Learning Models

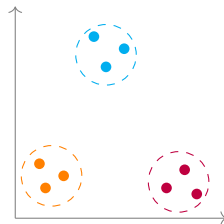
Classification (Supervised)



Regression (Supervised)



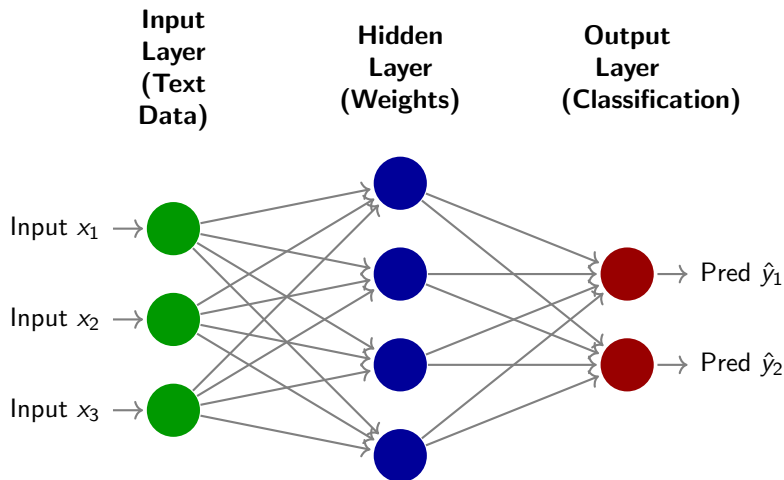
Clustering (Unsupervised)



Why Neural Networks?

- **The Limitation of Standard ML:** Traditional machine learning algorithms struggle with the immense complexity, sequential nature, and nuance of human language.
- **The Neural Advantage:** Neural networks excel at Natural Language Processing because they can automatically discover hidden, non-linear relationships.
- **Abstract Features:** Deep, stacked layers inside the network mathematically transform raw text vectors into deep semantic representations without requiring manual feature extraction.

Diagram: Neural Network Architecture



Why Not Traditional Models?

- **Linear Regression:** Designed exclusively to predict *continuous numerical values*. It mathematically cannot predict discrete SATS triage categories (Red, Orange, Yellow).
- **Logistic Regression:** While it predicts categories, it assumes inputs are linear and independent. It fails to capture complex syntactic dependencies.
- **The NLP Failure Point:** These methods rely on basic “Bag of Words” counts. They completely lose the sequence and context of language, failing to differentiate the profound semantic difference between “*my chest is NOT tight*” and “*my chest IS tight*”.

The Math of Neural Networks: Forward & Penalty

Every neural network executes a strict 4-step mathematical loop to learn.

1. The Forward Pass (The Guess)

$$y = Wx + b$$

Transforms independent input data (x) into a dependent prediction (y) using internal weights (W) and biases (b).

2. Cross-Entropy Loss (The Penalty)

$$\mathcal{L} = - \sum_{i=1}^C y_i \log(\hat{y}_i)$$

If the dependent output is wrong, this calculates the exact mathematical penalty. Because of the negative log, predicting the wrong triage category with high confidence results in a massive mathematical penalty ($\mathcal{L}$).

The Math of Neural Networks: Investigation & Correction

3. Backpropagation (The Investigation)

$$\frac{\partial \mathcal{L}}{\partial W} = \frac{\partial \mathcal{L}}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z} \cdot \frac{\partial z}{\partial W}$$

Uses the Chain Rule of calculus to travel backward through the network. It calculates the exact gradient (the slope of the error) to determine which specific weights caused the bad guess.

4. Optimization (The Correction)

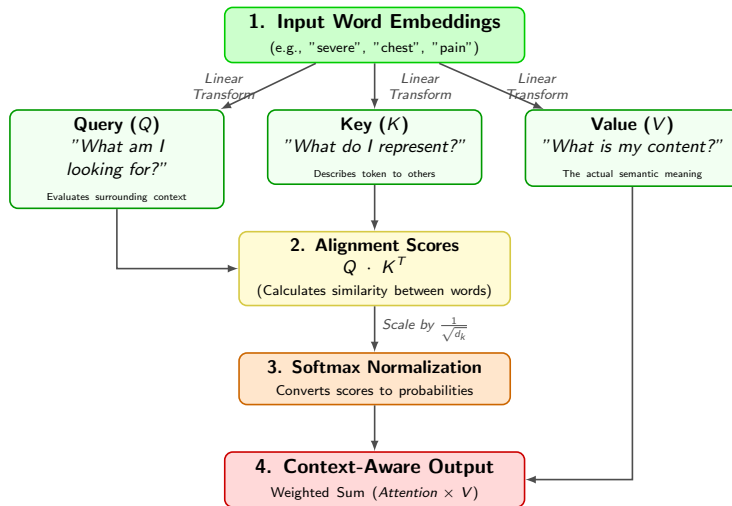
$$W_{new} = W_{old} - \alpha \left(\frac{\partial \mathcal{L}}{\partial W} \right)$$

Using Gradient Descent, the algorithm physically adjusts the weights (W) in the opposite direction of the gradient by a step size defined by the Learning Rate (α).

Transformers: Revolutionizing NLP

- **The Problem:** Standard ML and early neural networks struggled with long text because they read words sequentially (one by one).
- **The Breakthrough:** The 2017 "*Attention Is All You Need*" paper introduced the Transformer architecture.
- **Parallel Processing:** Transformers abandoned sequential reading, allowing the model to read and process entire sentences in parallel.

Diagram: The Self-Attention Mechanism



The Mathematics of Self-Attention

The self-attention matrix calculates exactly how much words relate to each other:

Self-Attention Formula

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V$$

- **Independent Variables:** Queries (Q), Keys (K), and Values (V).

Softmax Weighting

$$a_j = \frac{\exp(e_j)}{\sum \exp(e_k)}$$

- **Dependent Variable:** The final Attention Weight (a_j). It depends entirely on the raw mathematical scores of the surrounding words, mathematically assigning massive weight to critical triage keywords.

BERT: Bidirectional Encoder Representations

- **Bidirectionality:** Pre-machine learning models and early NLP read text strictly left-to-right. BERT reads context from both the left and the right simultaneously to deeply understand semantic meaning.
- **Pre-Training via MLM:** BERT learns the language through Masked Language Modeling.
- **The Process:** It randomly hides 15% of the words in a sentence and attempts to guess them based on the surrounding context.

BERT Pre-Training Mathematics

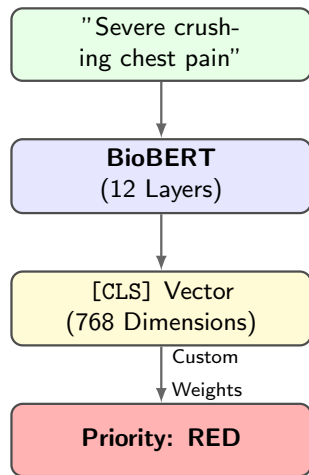
Masked Language Modeling Loss

$$\mathcal{L}_{LM} = - \sum \log P(t_i | H^{(L)}; \theta)$$

- **Independent Variables ($H^{(L)}$):** The surrounding, unmasked context words the model is allowed to read.
- **Dependent Variable ($\mathcal{L}_{LM}$):** The loss penalty. The penalty size depends entirely on the probability (P) the model assigned to the correct hidden word.

BioBERT & Triage Classification

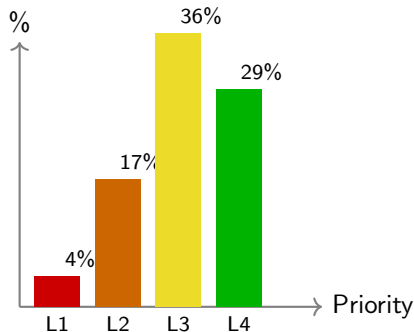
- **Training Data:** Pre-trained on millions of PubMed Biomedical Abstracts, providing deep, pre-existing intuition for medical jargon.
- **Processing:** Text is pushed through 12 neural layers to generate a compressed summary.
- **Independent Variable:** The [CLS] token—a 768-dimensional mathematical vector summarizing the complaint.
- **Dependent Variable:** The model multiplies this 768-dim vector by custom weights to predict the SATS triage category.



The Training Data: Sources & Statistical Profile

- **Data Provenance:** Pre-trained on 18B words from PubMed/PMC and Fine-tuned on the **Triagegeist Dataset** (synthetic, 80k records). This allows us to model ED statistics with **zero privacy risk**.
- **Statistical Distribution:**
 - **Mode (Most Frequent):** Level 3 (Yellow) at 36.15%.
 - **Median:** Level 3, representing a moderate baseline.
 - **Rarest Class:** Level 1 (Red) at just 4.03%.
- **The Imbalance Ratio:** There is a $\sim 9:1$ ratio between the mode (most frequent) and the most critical emergency class.

Distribution of 80,000 Records



How Data Statistics Shape Model Performance

- **Gradient Dominance:** Because the mode (Level 3) and Level 4 make up $\sim 65\%$ of the training data, the neural network's loss gradient naturally prioritizes learning these common cases.
- **Clinical Grounding (TEWS & SATS):** To overcome the risk of the model over-relying on text alone, the BioBERT training was mathematically anchored to objective TEWS and SATS labels.
 - *The Effect:* This gives the text embeddings "clinical intuition" by forcing them to correlate with actual physical vital signs.
- **Final Architecture:** By combining this TEWS-aligned medical triage model with a deterministic TEWS fallback system, we overcome statistical biases and ensure that critical, life-threatening cases are never missed.